

gdTAI V4.1-GPU Gate C

Nested cross-dataset evaluation under the precommitted purity and recall guardrails.

Complete, but no model is eligible. All three outer folds finished. No Stage-1 candidate satisfied the literal per-source NK-passage condition, and no fair comparator or raw legacy score satisfied either final operating mode. V3 Round 14 remains the promoted balanced default.

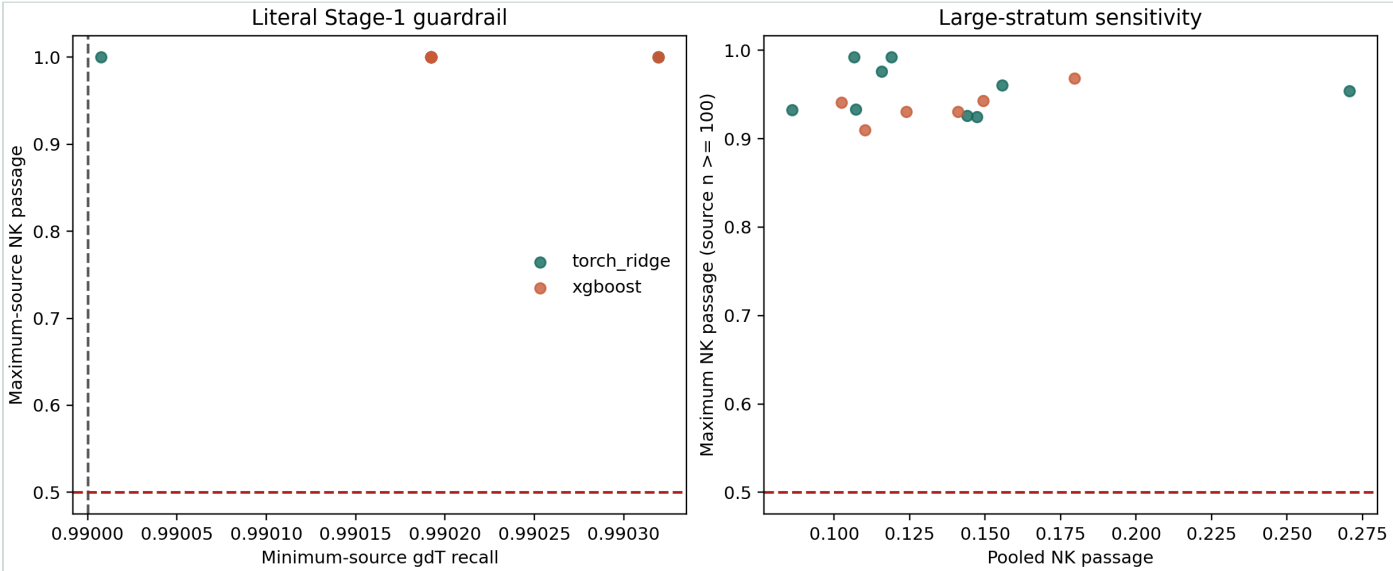
What Was Evaluated

Ground truth, 197 genes, grouped splits, CD4/Treg exclusions, calibration, and operating thresholds were unchanged. Stage 1 compared three deterministic GPU ridge penalties with two fixed GPU XGBoost configurations. Primary gdT and productive alpha-beta T recall had to remain at least 0.99 and 0.98 in every inner primary source, while strict-NK passage had to remain at most 0.50 in every NK source.

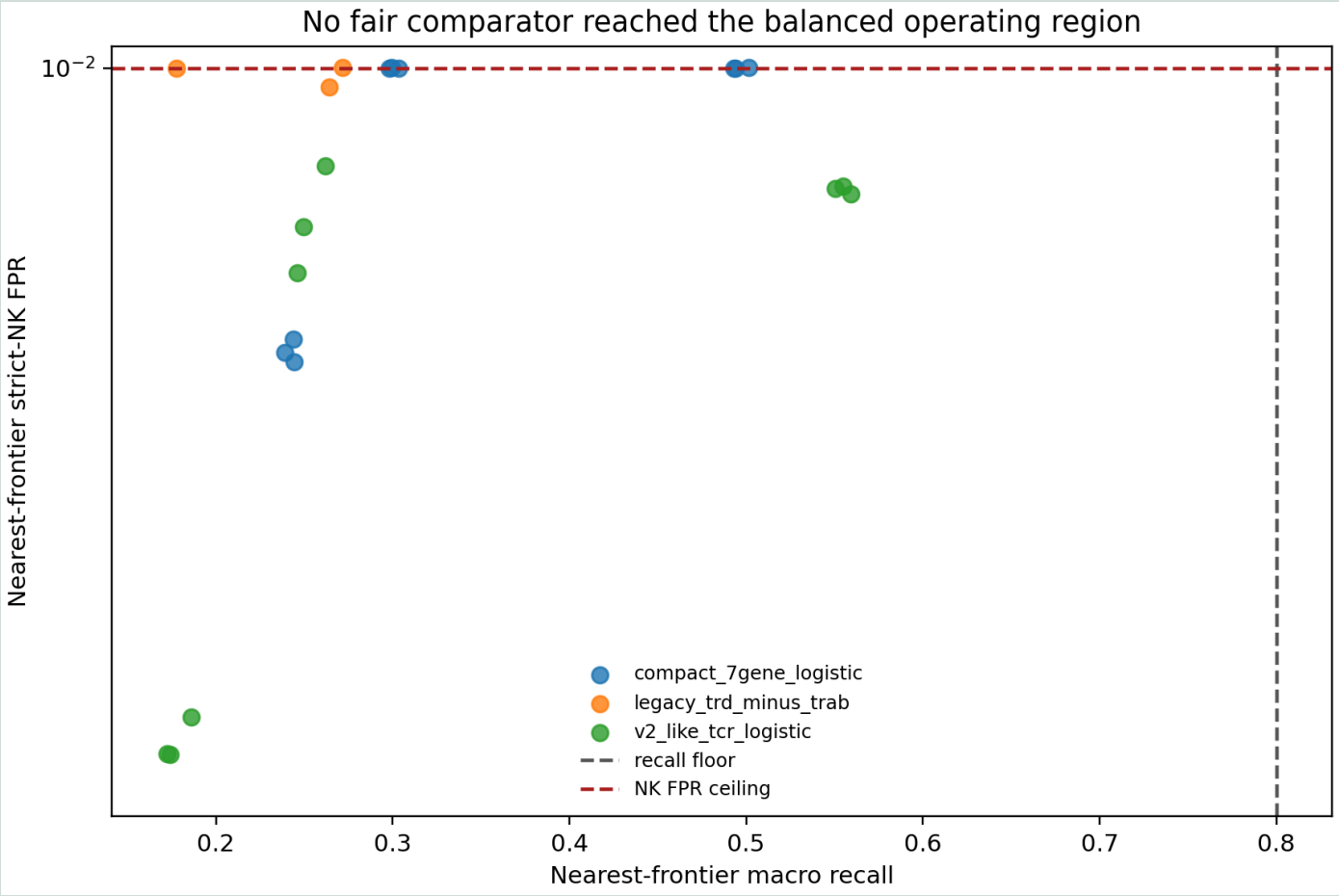
Silver, sorted, dual, ambiguous, and unlabeled cells never selected models or thresholds. Fair compact seven-gene and 153-gene TCR comparators used GPU ridge with the same grouped folds, exclusions, hard strict-NK negatives, calibration, and final thresholds. The raw TRD-minus-TRAB score was evaluated without refitting.

Stage-1 Blocking Result

All 15 candidate-by-outer-fold evaluations converged, but 0/15 passed. At the highest threshold retaining the required T-cell recalls, at least one NK source still had passage above 50%. The one-cell GSE190870 stratum makes the literal maximum-source statistic discontinuous, but it does not explain away the result: after restricting the diagnostic calculation to NK sources with at least 100 cells, the best worst-source passage still ranged from 90.97% to 93.23% across outer folds. Pooled NK passage was lower at 8.60% to 12.38%, showing marked source heterogeneity rather than adequate general separation. These post-hoc summaries do not alter the official failure.



Literal and sample-size-aware Stage-1 diagnostics.



Balanced comparator near-miss operating points.

outer_fold_id	family	minimum_source_gdt_recall	minimum_source_abt_recall	pooled_nk_passage	maximum_nk_passage_sources_nge_100	worst_nk_source_nge_100	worst_nk_source_nge_100_n	maximum_source_nk_passage	worst_nk_source_n
outer_0_HRA005041	xgboost	0.9903	0.9896	0.1238	0.9305	GSE287301	1539	1	1
outer_0_HRA005041	xgboost	0.9903	0.9903	0.1409	0.9305	GSE287301	1539	1	1
outer_1_GSE144469	xgboost	0.9902	0.9989	0.1025	0.9409	BALF_BLOOD_COPD	254	1	1
outer_1_GSE144469	torch_ridge	0.9902	0.9984	0.1066	0.9921	BALF_BLOOD_COPD	254	1	1
outer_2_BALF_BLOOD_COPD	torch_ridge	0.9902	0.9992	0.08601	0.9323	GSE144469	7804	1	1
outer_2_BALF_BLOOD_COPD	torch_ridge	0.9902	0.9992	0.1072	0.9334	GSE144469	7804	1	1

Fair Comparator Results

No compact seven-gene ridge, V2-like TCR-gene ridge, or raw legacy-score threshold met the balanced or high-purity constraints. Therefore there are no valid held-out predictions from which to claim F1, calculate a paired bootstrap advantage, or select a V4 model.

outer_fold_id	model_id	nearest_objective	nearest_recall	nearest_paired_abt_fpr	nearest_strict_nk_fpr	recall_requirement	paired_abt_fpr_limit	strict_nk_fpr_limit
outer_0_HRA005041	compact_7gene_logistic	0.3681	0.2441	0.002009	0.006741	0.8	0.002	0.01
outer_0_HRA005041	legacy_trd_minus_trab	0.3939	0.2643	0.001998	0.009731	0.8	0.002	0.01
outer_0_HRA005041	v2_like_tcr_logistic	0.2734	0.1862	0.002009	0.003888	0.8	0.002	0.01
outer_1_GSE144469	compact_7gene_logistic	0.4221	0.3033	0.0009419	0.009992	0.8	0.002	0.01
outer_1_GSE144469	legacy_trd_minus_trab	0.2996	0.1777	0.0001199	0.01	0.8	0.002	0.01
outer_1_GSE144469	v2_like_tcr_logistic	0.3402	0.262	0.001997	0.008672	0.8	0.002	0.01
outer_2_BALF_BLOOD_COPD	compact_7gene_logistic	0.6355	0.5014	0.001144	0.01	0.8	0.002	0.01
outer_2_BALF_BLOOD_COPD	legacy_trd_minus_trab	0.4002	0.2715	0.0004062	0.01001	0.8	0.002	0.01
outer_2_BALF_BLOOD_COPD	v2_like_tcr_logistic	0.6615	0.5595	0.001999	0.008323	0.8	0.002	0.01

What Is Not Estimable

V4 Stage 2 was not entered because Stage 1 had no eligible candidate. Extension-cohort model false-positive rates and paired hierarchical-bootstrap differences are consequently not estimable. This is a valid negative Gate C result, not missing evidence and not authorization to relax a threshold after observing outcomes.

Interpretation

The precommitted Stage-1 rule is statistically brittle because it takes a hard maximum over NK strata ranging from one cell to more than 90,000 cells. However, the ≥ 100 -cell sensitivity analysis shows a second, substantive problem: at the recall-preserving threshold, at least one well-populated dataset passes more than 90% of strict NK cells. Thus V4.1 failed because T-lineage sensitivity and cross-dataset NK rejection were incompatible under this feature panel, not merely because of a one-cell denominator. A future V4.2 may use hierarchical or minimum-stratum guardrails, but it must also improve source-robust discrimination and be declared before re-evaluation.

Decision Boundary

No V4.1 model is promoted or released. No whole-atlas inference is permitted. V3 Round 14 remains the balanced default pending a separately reviewed redesign.